

What Is Machine Learning?

Machine learning (ML) is a branch of artificial intelligence that gives computer systems the ability to learn from data and improve their performance on tasks without being explicitly programmed. Instead of following hand-crafted rules, an ML model finds patterns in historical data and uses those patterns to make predictions or decisions on new, unseen data. The field was formally named by Arthur Samuel in 1959, who defined it as the study that gives computers the ability to learn without being explicitly programmed.

Types of Machine Learning

Machine learning is broadly divided into three paradigms. Supervised learning is the most common: the model is trained on labelled examples, where both the input features and the correct output are provided. Common supervised tasks include classification (predicting a category) and regression (predicting a continuous value). Examples include email spam detection, house price prediction, and image classification. Unsupervised learning works with unlabelled data. The model discovers hidden structure on its own. Clustering algorithms such as K-Means group similar data points together. Dimensionality reduction techniques such as Principal Component Analysis (PCA) compress data while preserving the most important variance. Reinforcement learning trains an agent to make sequential decisions by rewarding desired behaviour and penalising undesired behaviour. The agent interacts with an environment, observes state, takes actions, and receives a reward signal. This paradigm powers game-playing systems like AlphaGo and robotics control systems.

Key Terminology

Features are the measurable input variables used to make a prediction; they form the columns of a dataset. Labels are the target outputs the model learns to predict. A dataset is split into training data, used to fit the model, and test data, used to evaluate it on unseen examples. Overfitting occurs when a model memorises the training data so well that it performs poorly on new data. Underfitting occurs when a model is too simple to capture the underlying patterns. The bias-variance tradeoff describes the tension between a model that is too rigid (high bias) and one that is too sensitive to training noise (high variance). Hyperparameters are configuration settings chosen before training begins, such as learning rate, number of trees in a random forest, or the depth of a neural network.

Common Algorithms

Linear Regression models the relationship between a continuous target variable and one or more input features by fitting a straight line (or hyperplane) that minimises the sum of squared errors. It is interpretable and computationally cheap but assumes a linear relationship. Logistic Regression, despite its name, is a classification algorithm. It outputs the probability that an input belongs to a given class using the sigmoid function, thresholded at 0.5 for binary classification. Decision Trees split the feature space into regions by asking a series of yes/no questions. Each internal node represents a feature test; each leaf represents a predicted output. Random Forests build many decision trees on random subsets of the data and features, then average their predictions. This ensemble method reduces variance and is robust to overfitting. Support Vector Machines (SVM) find the hyperplane that maximises the margin between classes. Using the kernel trick, SVMs can separate data that is not linearly separable by projecting it into a higher-dimensional space.

Neural Networks and Deep Learning

A neural network is a function approximator loosely inspired by biological neurons. It consists of an input layer, one or more hidden layers, and an output layer. Each connection has a weight; each neuron applies a non-linear activation function such as ReLU, sigmoid, or tanh to a weighted sum of its inputs. Training uses backpropagation: the gradient of the loss with respect to each weight is computed via the chain rule and the weights are updated using gradient descent. Deep learning refers to neural networks with many hidden layers. Convolutional Neural Networks (CNNs) use spatial filters to process images and have achieved human-level performance on image recognition benchmarks. Recurrent Neural Networks (RNNs) process sequential data by maintaining a hidden state that captures context from previous time steps. Transformer architectures, introduced in the paper 'Attention Is All You Need' (2017), replaced recurrence with self-attention mechanisms and became the foundation for large language models such as GPT and BERT.

Model Evaluation and Validation

Evaluating a model correctly is as important as training it. For classification tasks the primary metrics are accuracy (fraction of correct predictions), precision (fraction of positive predictions that are truly positive), recall (fraction of true positives that were detected), and the F1 score (harmonic mean of precision and recall). The ROC curve plots the true positive rate against the false positive rate at varying thresholds; the area under the curve (AUC-ROC) summarises overall discriminative power. For regression tasks, common metrics include Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE). K-fold cross-validation partitions the dataset into K equal folds, trains on K-1 folds, and evaluates on the remaining fold, rotating K times. This gives a more reliable performance estimate than a single train/test split and helps detect overfitting.